# Parameter-Efficient Transformer Fine-tuning for Bilingual Toxic Text Classification & Regression

LoRA Adapters vs Full Fine-tuning — Quality · Speed · Robustness on a Noisy RU/EN Corpus

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## ① SETUP — QUESTION, DATA, MODELS

### THE QUESTION

**Case Study 2.6 — Parameter-Efficient Transformer Fine-tuning.** Take a small pretrained transformer LM, apply LoRA layers to specialize it for a niche domain corpus, and document the trade-offs between adaptation speed, GPU memory footprint, and downstream perplexity. NLP course case study at Innopolis University. Topic overlaps with the **Generative AI** course — Alexander Malyy, a fellow Generative AI student, assisted with select tasks.

**Quality**
F1 on subtle, context-dependent toxic content

**Speed**
Training cost and inference latency

**Robustness**
l33tspeak, char spacing, homoglyphs

**Can LoRA — training only 1% of parameters — match full fine-tuning on all three axes?** We answer this with two independent experiments:

**Exp 1 — Classification (sections ②④):** DistilBERT full FT vs LoRA vs TF-IDF on toxic-text detection. Metrics: F1, EN/RU gap, obfuscation robustness, class-imbalance ablations.

**Exp 2 — Autoregressive LM (section ③):** DistilGPT-2 full FT vs LoRA rank sweep (r=4/8/16). Metrics: perplexity, training time, memory, checkpoint size.

### DATASET

| 460K | 4 | 92K | 2 |
|---|---|---|---|
| samples after LSH | Kaggle datasets | test set seed=42 | languages RU + EN |

Russian 56% | English 44%

Safe 80% | Toxic 20%

HTML/URLs stripped. MinHash-LSH t=0.80 removed near-duplicates. One train/val/test split (seed=42) shared by all models.

### MODELS & SETUP

Exp 1 — Classification

| Model | Params | Data |
|---|---|---|
| TF-IDF + LogReg | 10K feat | 133K bal. |
| DistilBERT full FT | 67.6M | 331K |
| DistilBERT + LoRA r=8 | 739K (1.09%) | 90K bal. |

LoRA: α=16, q/v projection. Merged before inference → serving cost = full FT.

Ablations (Exp 1)

| # | Question | Change vs baseline |
|---|---|---|
| 1 | Does class balancing help? | All 3 models, undersample majority |
| 2 | Does weighted loss help? | DistilBERT, 4× toxic weight |
| 3 | Data size vs architecture? | LoRA vs full FT, both at 90K |
| 4 | Robust to obfuscation? | All 3 + rule-based deobfuscation |

Exp 2 — Autoregressive LM

| Model | Params | Data |
|---|---|---|
| DistilGPT-2 full FT | 82M (100%) | 90K |
| DistilGPT-2 + LoRA r=4/8/16 | 73–295K | 90K |

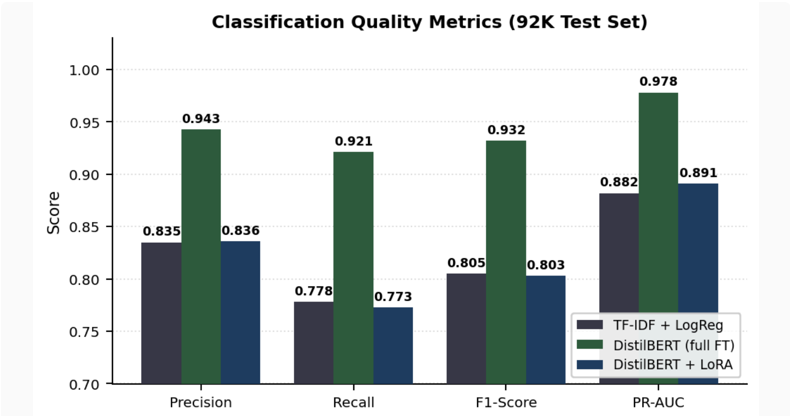
Rank sweep: same data, 1 epoch. Metrics: PPL, time, memory, checkpoint size.

### CODE & REPRODUCE

Full implementation, training scripts, and experiment logs:
github.com/ArthurBabkin/GenAI-Safety-Filter/tree/nlp-case-study

## ② EXP 1: CLASSIFICATION — DOES LORA MATCH FULL FT ON F1, SPEED, ROBUSTNESS?

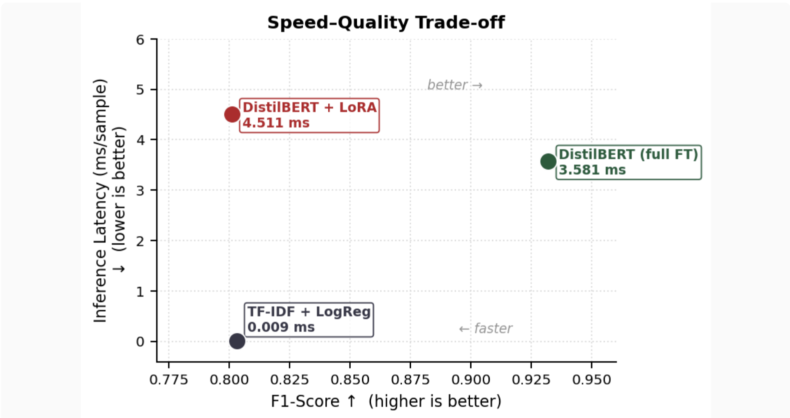
### HEADLINE RESULTS (92K TEST SET)




| Model | F1 | Precision | Recall | Latency | Throughput |
|---|---|---|---|---|---|
| TF-IDF + LogReg | 0.805 | 0.835 | 0.778 | **0.009 ms** | **112K/s** |
| **DistilBERT full FT** | **0.932** | **0.943** | **0.921** | 3.58 ms | 287/s |
| DistilBERT + LoRA | 0.804 | 0.836 | 0.773 | 4.51 ms | 226/s |

**Answer: No.** Full FT leads by **+12.8pp F1.** LoRA has transformer-class latency (4.5 ms) at LogReg-class quality (F1 0.80) — worst-of-both-worlds for serving. Its wins are in training only.

### SPEED VS QUALITY TRADE-OFF




LoRA occupies the worst quadrant: transformer-class latency (4.5 ms) at LogReg-class quality (F1 0.80). Training benefits don't survive into serving.

### PER-LANGUAGE: EN VS RU

| Model | EN F1 | RU F1 | Gap |
|---|---|---|---|
| TF-IDF + LogReg | 0.828 | 0.785 | 4.3pp |
| **DistilBERT** | **0.936** | **0.928** | **0.9pp** |
| LoRA | 0.849 | 0.761 | 8.8pp |

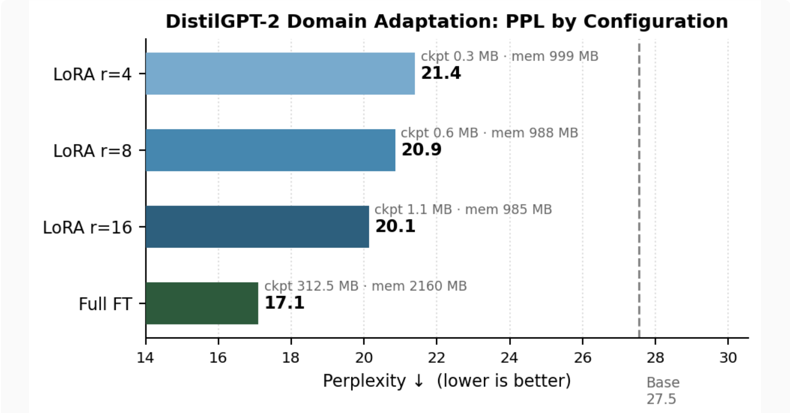
Full FT (67.6M trainable params) realigns the English-only backbone for Cyrillic. LoRA's 1% parameter budget is insufficient — the English bias leaks through, yielding the largest EN/RU gap of all three models.

## ③ EXP 2: AUTOREGRESSIVE LM — DISTILGPT-2 RANK SWEEP: SPEED · MEMORY · PERPLEXITY

### PERPLEXITY RESULTS BY RANK

DistilGPT-2 (82M) · 90K texts · 1 epoch · AdamW + linear warmup · batch 16 · Apple MPS

| Config | Trainable | PPL ↓ | ΔPPL | Time | Mem MB | Ckpt MB |
|---|---|---|---|---|---|---|
| Base (no FT) | — | 27.5 | — | — | — | — |
| LoRA r=4 | 73K | 21.4 | +6.1 | 30 min | 999 | 0.3 |
| LoRA r=8 | 147K | 20.9 | +6.7 | 33 min | 988 | 0.6 |
| LoRA r=16 | 295K | 20.1 | +7.4 | 35 min | 985 | 1.1 |
| **Full FT** | **82M (100%)** | **17.1** | **+10.4** | **43 min** | **2160** | **312** |




### THREE TRADE-OFFS EXPLAINED

**Speed**
All LoRA ranks finish in ~30 min. Full FT: 43 min. Rank has *no effect* on speed — r=4 ≈ r=16.

**Memory**
LoRA ~990 MB (any rank). Full FT: 2160 MB.
**2.2× savings**
— memory is rank-independent.

**Perplexity**
Rank *does* improve PPL: r=4→21.4, r=16→20.1. Full FT: 17.1. Rank is the quality knob.

**Checkpoint**
LoRA r=16: 1.1 MB vs 312 MB full FT.
**285× smaller**
— key for multi-variant deployment.

**Insight:** LoRA decouples quality from cost. Raise rank to lower perplexity with no speed or memory penalty. Full FT wins on quality but requires 2× memory and a 285× larger checkpoint.

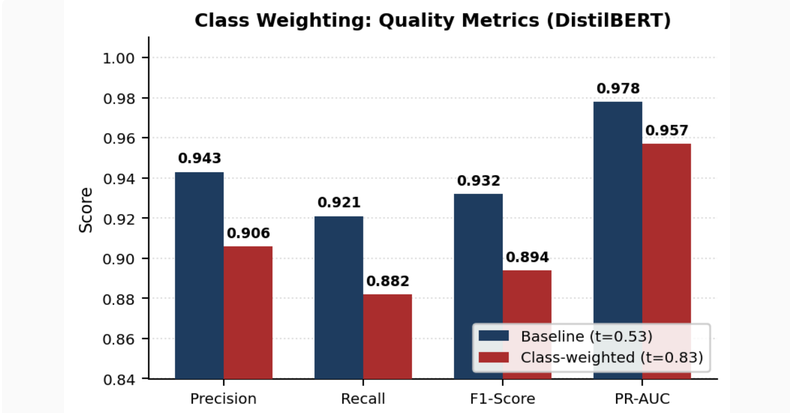
## ④ EXP 1 DEEP DIVE — CLASSIFICATION ABLATIONS → JOINT CONCLUSIONS

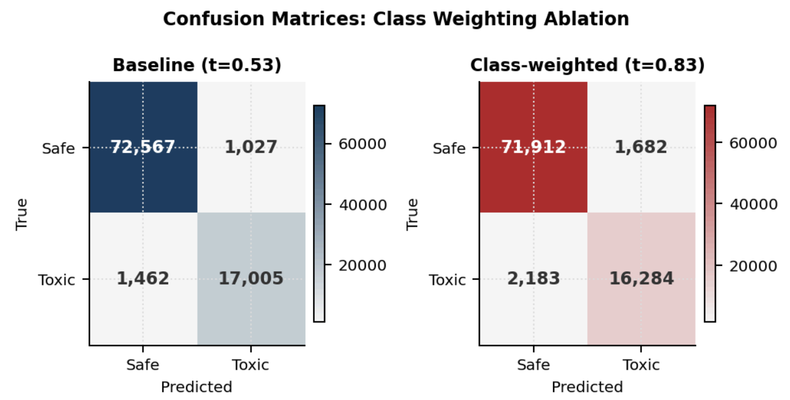
### ABL. 1: CLASS IMBALANCE

| Model | Baseline | Balanced | Δ |
|---|---|---|---|
| **LogReg** | 0.765 | **0.805** | **+4.1pp** |
| DistilBERT | **0.932** | 0.884 | −4.7pp |
| **LoRA (same size)** | 0.793 | **0.804** | **+1.1pp** |

Effect is model-capacity dependent: linear models gain from balanced priors (+4.1pp), while pretrained transformers on 331K show no majority-class bias — undersampling only removes useful training signal (−4.7pp).

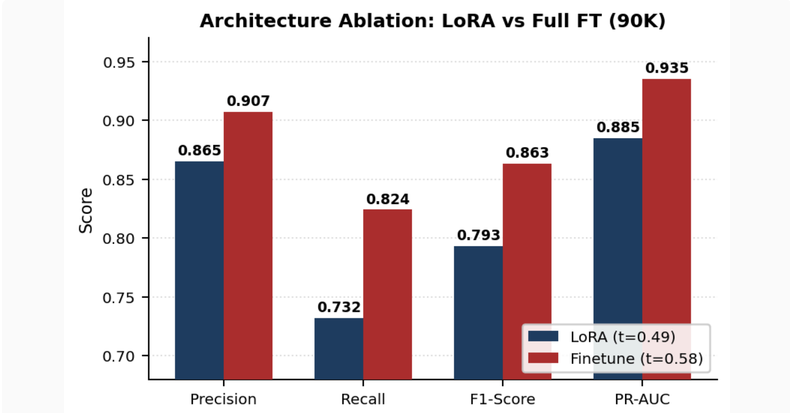
### ABL. 2: CLASS WEIGHTING

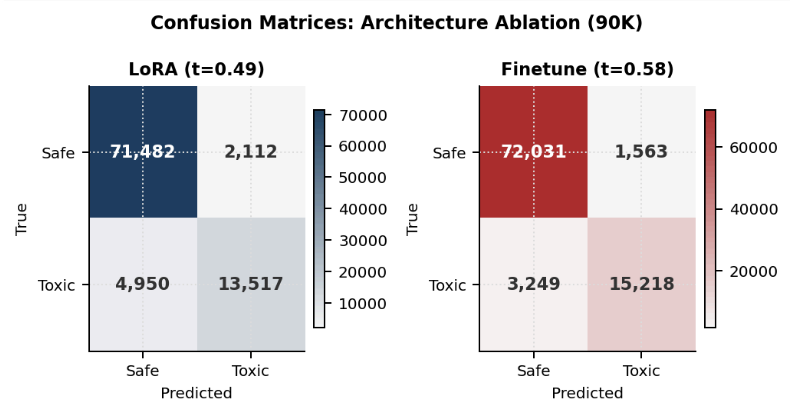



4× toxic weight: F1 **−3.8pp**, PR-AUC −2.1pp, threshold shifts 0.53→0.83. DistilBERT already achieves 92% recall on unweighted data — loss reweighting assumes a bias that doesn't exist, inflating toxic-class probabilities and breaking calibration.

### ABL. 3: ARCHITECTURE VS DATA SIZE

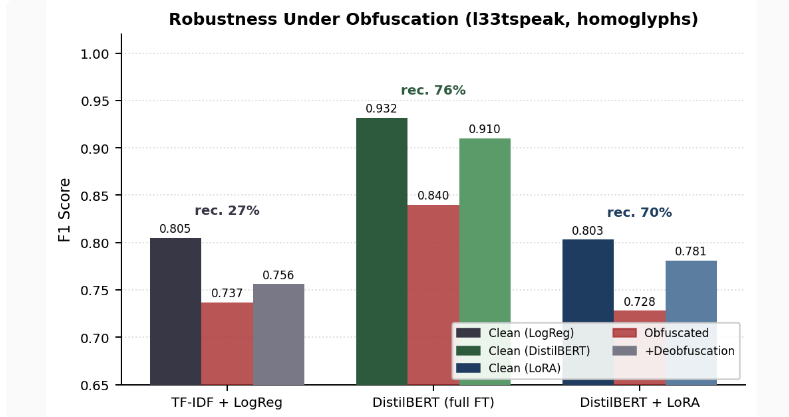



| Model | Recall | F1 | PR-AUC |
|---|---|---|---|
| LoRA 90K | 0.732 | 0.793 | 0.885 |
| **Full FT 90K** | **0.824** | **0.864** | **0.935** |
| Δ (arch.) | +0.092 | **+7.1pp** | +0.050 |

**13pp total gap decomposes evenly: ~7pp architecture + ~7pp data volume.** Recall is hit hardest (+9.2pp gap), showing LoRA's constrained parameter budget limits learning of minority-class discriminative features.

### ABL. 4: ROBUSTNESS UNDER OBFUSCATION




| Model | Clean | Obf. | +Deobf. | Recov. |
|---|---|---|---|---|
| LogReg | 0.805 | 0.737 | 0.756 | 26.8% |
| **DistilBERT** | **0.932** | **0.840** | **0.910** | **75.8%** |
| LoRA | 0.804 | 0.728 | 0.781 | 69.5% |

**Zero retraining needed.** A rule-based deobfuscation preprocessor recovers 76% of DistilBERT's obfuscation drop — high-value defense against evasion. TF-IDF has no char-level representation: only 27% recovery regardless.

### CONCLUSIONS

Exp 1 — Classification

**Full FT wins on quality:** F1 0.932, EN/RU gap 0.9pp, 75.8% robustness recovery. Best when compute allows.

**LoRA = worst-of-both for serving:** transformer latency (4.5 ms) at LogReg-class F1 (0.80). Benefits only during training.

**Free robustness:** rule-based deobfuscation recovers 76% of transformer drop — zero retraining.

Exp 2 — Autoregressive LM

**LoRA rank = quality dial:** r=4→21.4 PPL, r=16→20.1 PPL — no speed or memory penalty. Rank is the only quality knob.

**Resource savings are rank-independent:** 2.2× less memory, 285× smaller checkpoint vs full FT. Decisive for multi-variant deployment.

Joint Takeaway

**LoRA's 1% parameter budget has a hidden multilingual cost:** EN/RU gap 8.8pp vs 0.9pp for full FT — frozen English-biased weights can't be realigned without full gradient flow. Critical for any non-English deployment.

**Use LoRA when:** rapid prototyping, compute-constrained training, or serving many domain variants from one base. **Use full FT when:** production quality, multilingual corpus, or maximum recall is required.

### LIMITATIONS & FUTURE WORK

**Domain mismatch.** Social-media comments only. LLM-output moderation unverified — different register and length distribution.

**English-centric backbone.** XLM-R or mDeBERTa would close the EN/RU gap by design; distilbert-base-uncased was chosen for speed.

**Single-seed training.** No variance across random seeds; small numeric deltas are indicative only.

**Adversarial scope.** 4 rule-based obfuscation families (leetspeak, spacing, repetition, mixed case). No paraphrase or semantic attacks tested.

**Binary label collapse.** Severity level, target group, and context discarded; all 4 source datasets reduced to safe/toxic.

**Regression not evaluated.** Toxicity severity as a continuous score (ordinal regression / MSE objective) was not explored — binary framing may discard signal useful for ranked moderation queues.

**Rule-based deobfuscation.** Brittle to novel evasion patterns; a learned character-level normalizer is natural future work.